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1. Apriori Algorithm - online shopping Transaction

Given dataset

Order	product
1	mobile earphones
2	mobile, charger
3	mobile phones, charger
4	mobile, earphones
5	mobile earphones

Assume minimum support count = 3

pseudocode:

Apriori (Transaction - Dataset, min support)

1. C1 - generate all 1-itemsets
2. count support of every item in C1
3. L1 ← remove items whose support < min-support
4. k ← 2
5. while L(k-1) is not empty
   ck ← generate candidates from L(k-1)
   for each candidate c in ck:
   If any (k-1) - subset of c is not in L(k-1).
   Remove c
   count support of remaining candidates

1kE candidates having support ≥

min - support

k ← k+1

6. Return all frequence itemsets $L_1, L_2 \ldots$

2. FP - Growth Library Book Borrowing

Dataset:

Borrow ID	Book
B1	Data science, python
B2	python, machine learning
B3	Data science, machine learning
B4	Data science, python, machine learning
B5	python, machine learning.

minimum support count = 2

Pseudocode:

FP - Growth (D, min - support)

1. Scan D and calculate support of each item
2. Remove items whose support < min-support
3. Sort remaining items according to decreasing support
4. Create an empty FP Tree.
5. For each infrequent items.

Romove infrequent items
Sort-items T by frequency
update node Count

6. Create header table linking identical item
7. mine the FP-Tree pattern bases
construct Conditional FP-Trees
Generate Frequence itemsets
8. Return all frequent items.

Constraint-Based Frequent itemset mining

Dataset

Student	Courses
S1	python, Database
S2	python, AI
S3	Database, AI
S4	python, Database, AI
S5	python

Final Frequent Itemsets

{python}

{python, database}

{python, AI}

Pseudocode:

1. Define constraint
2. Every itemset must contain "python"
3. Generate support q
4. count support
5. L_1 ∈ Candidates satisfying min-support
6. for k=2 while satisfying min-support

Generate candidates itemset from(k-1)

for each candidate c:

If "python" is not in c

prune c

else

Add c to Lk

7. Return All Lk

3. Association rule Generation:

Dataset

Itemset	support count
{Internet}	4
{streaming}	4
{cloud, streaming}	3
{streaming, cloud, storage}	3
	2

Assume minimum confidence = 60%.

$$\text{Confidence}(A \to B) = \frac{\text{support}(A \cup B)}{\text{support}(A)} \times 100$$

Pseudo code:

GENERATE - RULES (Frequent - Items min - confidence)

1. For each frequent itemset F
2. Generate all non-empty subsets A of F
3. B ← F - A
4. confidence ← support (F) / support (A)
5. If confidence ≥ min - confidence
6. Generate ≥ min - confidence.
7. Accept the rule
8. else
   Reject the rule:
9. RETURN all accepted rule.

4. multilevel Association Rule mining

Dataset

Patient	Treatment
P1	Health care, Cardiology ECG
P2	Healthcare, orthopedics
P3	Healthcare, cardilogy
P4	Health, cardiology ECG
P5	Healthcare, orthopedics

Pseudocode:

1. Multilevel - Association - minimung (D, Hierany)

→ Define the Treatment Hierarchy

→ start from the highest / general level

→ For each hierachy level:

extract items belonging to that level

count support of each item / itemset

→ Remove items below min- support

Generate association true rule frequence items

Calculate confidence for each rule

→ move to the next, more detailed hierarchy level

→ Repeat the mining process

→ Return frequent patterns & rule from all levels